INCH-POUND

MIL-G-82841(OS)
5 February 1990
SUPERSEDING
WS 21055A
28 August 1986
(See 6.5)

MILITARY SPECIFICATION

GRAPHITE, ATJS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of ATJS graphite, referred to herein as the graphite (see 6.1.)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

MIL-STD-453 Inspection, Radiographic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 9620

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

WR-43 Preparation of Quality Assurance Provisions (Including Classification of Characteristics).

(Copies of WR-43 are available from the Commanding Officer, Naval Publications and Forms Center 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2.).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C 559 Bulk Density by Physical Measurement of Manufactured Carbon and Graphite Articles

ASTM C 561 Ash in a Graphite Sample

ASTM C 695 Compression Strength of Carbon and Graphite

ASTM C 714 Thermal Diffusivity of Carbon and Graphite by a Thermal Pulse Method

ASTM E 4 Standard Practices for Load Verification of Testing Machines

ASTM E 6 Definition of Terms Relating to Methods of Mechanical Testing

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ASTM E 9 Compression Testing of Metallic Materials At Room Temperature (DoD adopted)

ASTM E 83 Verification and Classification of Extensometers (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the graphite shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 **Form.** The graphite shall be molded using fine grained carbon-graphite in the form of cylindrical billets and processed to a minimum temperature of 4500 degrees Fahrenheit (^{o}F).

3.3 **Chemical and physical properties.** Chemical and physical properties of the graphite shall conform to the requirements specified in table I.

3.4 **Subsurface conditions.** The subsurface condition of the graphite shall be free from voids, inclusions and cracks.

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TABLE I. Chemical and physical properties.

Property	Value	
	Minimum	Maximum
Ultimate tensile strength, pounds per square inch (psi)		
With grain	4400	---
Across grain	3600	---
Tensile elastic modulus ($x10^6$ psi)		
With grain	---	2.0
Across grain	---	1.5
Tensile strain at failure, inch/inch (in/in)		
With grain	0.004	
Across grain	0.004	
Ultimate compressive strength (psi)		
With grain	10,000	---
Across grain	10,000	---
Compressive elastic modulus ($x10^6$ psi)		
With grain	---	1.20
Across grain	---	0.80
Thermal conductivity, BTU-inch/(hr)(ft)(°F) at 1600°F		
With grain	---	600
Across grain	---	500
Thermal expansion from 75 to 1800°F (in/in x 10^{-3})		
With grain	---	4.0
Across grain	---	5.0
Ash Content, Percent	---	0.06
Resistivity, ohm-cm x 10^{-4}	---	---
With grain	---	10
Density, gram/cubic centimeter (gm/cc)	1.80	---

3.5 **Billet dimensions.** The billets shall have a minimum diameter of 6.0 inches and a length of 8 to 12 inches.

3.6 **Shelf life.** The graphite furnished under this specification shall have an unlimited shelf life when stored under ambient conditions.

3.7 **Workmanship.** The graphite shall be uniform in appearance and free from foreign materials, cracks, voids or blemishes.

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4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, not does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the tests specified in table II and the quality conformance inspection specified in 4.4. Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination, tests, and inspection specified in table III.

4.4.1 Sampling for inspection.

4.4.1.1 Lot. For inspection purposes, a lot shall consist of a maximum of 150 billets. The lot shall be produced from one continuous production run manufactured from the same lot of material in one furnace run and subject to inspection at the same time.

4.4.1.2 Sample size. All billets shall be subjected to visual examination and radiographic inspection (see table III). Sampling for the remainder of the tests in table III shall be in accordance with table IV. A test ring shall be removed from each billet sampled. The ring shall be identified with the stock and lot number. Sufficient material shall be removed to obtain three test specimens for each test.

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TABLE II. First article inspection.

Tests	Requirement	Test Method
Tensile properties test		
Ultimate strength with and across grain	3.3	4.5.2.7
Strain to failure with and across grain	3.3	4.5.2.8
Elastic modulus with and across grain	3.3	4.5.2.9
Thermal conductivity test At 1600°F with and across grain	3.3	4.5.6
Thermal expansion test From 75 to 1800°F with and across grain	3.3	4.5.7

TABLE III. Quality conformance inspection.

Examination and Tests	Requirement	Test Method
Visual examination	3.5, 3.7, and section 5	4.5.1
Compressive properties test		
Ultimate strength With and across grain	3.3	4.5.3.7
Elastic modulus With and across grain	3.3	4.5.3.8
Density test	3.3	4.5.4
Ash content test	3.3	4.5.5
Resistivity with grain test	3.3	4.5.8
Radiographic inspection	3.4	4.5.9

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TABLE IV. Sampling for inspection.

Number of containers in the lot	Number of sample containers to be selected for inspection
1-3	ALL
4-15	3
16-25	4
26-40	5
41-65	7
66-110	10
111-150	15

4.4.2 Acceptance criteria. All billets from one lot shall be acceptable if each individual billet meets the requirements specified in 3.4, and the density and average property values determined from three tests of each billet tested in accordance with the sampling defined in 4.4.1 meets the requirements of table I. Record average values. If the lot is deemed unacceptable by the sampling of 4.4.1, each billet of the lot shall be subjected to the tests of table III for the failed physical property or properties only, and accepted on an individual basis.

4.4.3 Classification of characteristics. The characteristics verified by the inspections herein are classified as major or minor in accordance with WR-43. Inspections that verify major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the inspection. Inspections which are not annotated with a classification code and serial number are classified minor.

4.5 Methods of inspection. The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 Visual examination. Each sample of material shall be visually examine for conformance to 3.5, 3.7 and section 5. Contractor certification verifying conformance to 3.2 and 3.4 shall be required (see 6.2).

4.5.2 Tensile properties test.

4.5.2.1 Tension testing definitions. The definitions provided by ASTM E 6 shall apply to the terms used in this method of tension testing.

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4.5.2.2 **Tension testing machine.** The machine used for tension testing shall conform to the requirements of ASTM E 4. The testing machine shall have a capacity such that the breaking load of the test specimen falls between 10 and 90 percent of the scale capacity.

4.5.2.3 **Gripping devices.** Grips shall have self-aligning features in the spherical surface joint and shall always have a light coat of molybdenum disulfide base (Molikote or equivalent) lubricant to minimize the parasitic strains due to bending. The gripping devices shall be checked for parasitic strains by attaching strain gages at randomly selected 0, 90, 180, and 270 degree orientations in a planar area around the reduced specimen diameter and in the middle of the gage length. The total strain at each gage shall not vary at any load level from the average strain by more than 5 percent. A parasitic strain check of gripping devices shall be conducted periodically, at least once each month. A standard graphite specimen shall be established and nondestructively employed for checking purposes.

4.5.2.4 **Extensometers.** Extensometers shall comply with the requirements for the specific class described in ASTM E 83. Utilization of strain gages for measurement of strain in tension may be allowed upon approval of the procuring activity.

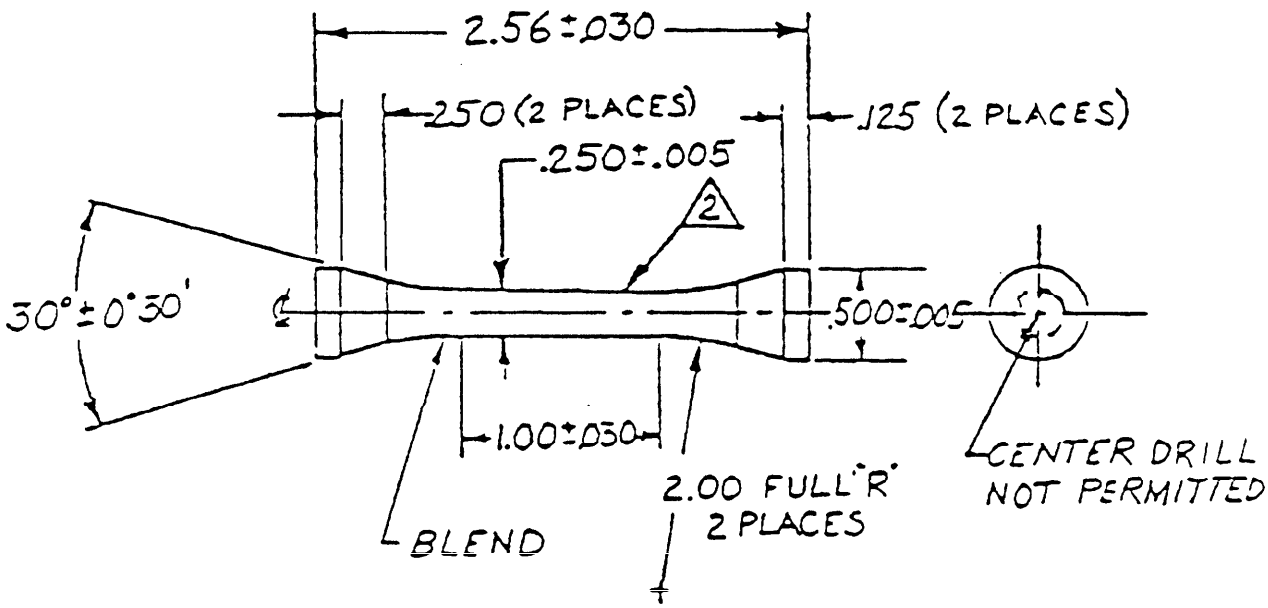
4.5.2.5 **Tension test specimens.** Tension test specimens shall be prepared as shown on figure 1. Adhesive film bands shall be applied around the reduced diameter at each end of the gage length and 0.937 $\pm$ 0.187 inch apart. Each band shall be 0.002 $\pm$ 0.0004 inch thickness and 0.187 inch width. After drying, the extensometer knife edges shall be mounted on the bands to prevent notches which could propagate failures.

4.5.2.6 **Speed of tension testing.** The test machine shall be calibrated at a free-running crosshead speed of 0.020 $\pm$ 10 percent/inch (minimum) or when rate of stressing of specimen technique is used, the applied force shall be 200 $\pm$ 50 pounds/minute.

4.5.2.7 **Tensile strength (M101).** The tensile strength shall be calculated by dividing the maximum load carried by the specimen during a tension test by the minimum cross-sectional area of the specimen. If the fracture takes place outside the uniform diameter gage length, the test shall be disregarded but recorded.

4.5.2.8 **Stress/strain to failure (M102).** Class B2 1-inch gage extensometers shall be used to obtain autographic records of the stress/strain curve. (Instron strain gage extensometer Model G-51-11 or equivalent.) The extensometer shall be kept on the tensile specimen through failure to measure the strain at ultimate strength of each specimen. The strain at failure shall be determined from the stress/strain graph to the nearest 1.0 percent. The strain values for each grain direction shall be averaged, using results only from those specimens used to derive the average ultimate strength.

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NOTE:

1. All diameters must be concentric within ± 0.001 T.I.R.

2. Gauge area to be hand polished with No. 000 dry paper until no circular grooves are visible to examination in accordance with 4.5.9. No scratches, tool marks or nicks are allowed in gauge area.

FIGURE 1. Tensile specimen.

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4.5.2.9 **Tensile elastic modulus (M103).** The elastic modulus shall be determined from the slope of the initial stress/strain curve generated on the machine strip chart or on the X-Y strain recorder. The elastic modulus value for each of both grain directions shall be averaged, using results only from those specimens used to derive the average ultimate strength.

4.5.3 **Compressive properties test.**

4.5.3.1 **Compression testing definitions.** Definitions of terms relating to compression testing in ASTM E 6 shall apply.

4.5.3.2 **Compression testing machine.** The machines used for compression testing shall conform to the requirements of ASTM E 4.

4.5.3.3 **Bearing blocks.** Spherical-seated bearing blocks or a sub-press, as recommended in ASTM E 9, shall be used to provide parallelism of the load strain.

4.5.3.4 **Deflectometers.** Deflectometers used for measuring strain in compression of graphite shall comply with the requirements for the specific class described in ASTM E 83.

4.5.3.5 **Compression test specimens.** The test specimens shall be 0.5 inch in diameter by 1.0 inch long.

4.5.3.6 **Speed of compression testing.** The test machine shall be calibrated at a free-running crosshead speed of 0.020 $\pm$ 10 percent/inch (minimum) or when rate of stressing of specimen technique is used, the applied force shall be 200 $\pm$ 50 pound/minimum.

4.5.3.7 **Ultimate compressive strength (M104).** Ultimate compressive strength shall be determined as defined in ASTM C 695.

4.5.3.8 **Compressive elastic modulus (M105).** The elastic modulus shall be determined from the initial slope of the stress/strain curve generated on the machine strip chart on the X-Y strain recorder.

4.5.4 **Density test (M106).** The density shall be determined in accordance with ASTM C 559.

4.5.5 **Ash content test (M107).** The ash content of the material shall be determined in accordance with ASTM C 561.

4.5.6 **Thermal conductivity (M108).** Thermal conductivity shall be determined in accordance with ASTM C 714 except a laser source shall be used.

4.5.7 **Thermal expansion test (M109).** Thermal expansion shall be determined using the twin telescope method and specimen dimensions of 1/2 by 1/2 by 2.5 inches long.

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4.5.8 **Resistivity test (M110).** The resistivity of the carbon-graphite material shall be calculated from the measured resistance, cross-sectional area, and the length of a machined specimen extracted in the with grain direction. Resistance shall be measured along the length of the specimen with a Kelvin Bridge ohmmeter. Suitable contacts shall be provided to connect the ends of the specimen, and a load shall be placed on the center of the specimen to assure contact of the potential points. Resistivity shall be calculated using the following equation:

$$\text{Resistivity (ohm-cm)} = \frac{R \times A}{L}$$

Where: R = resistance of specimen as determined in ohms.

A = cross-sectional area of specimen in square centimeters.

L = length of conductor in centimeters (distance between potential probes P1 and P2).

4.5.9 **Radiographic inspection (M111).** Each sample shall be radiographically inspected in accordance with MIL-STD-453. One radial X-ray shall be taken with the focal point centered along the axis of the sample. Dark and light exposures shall be taken as required.

4.6 **Inspection of packaging.** The graphite containers and container markings shall be visually examined for compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The graphite shall be packaged to afford adequate protection during shipment from the supply source to the first receiving activity and during storage under the shelf life period specified in 3.6. The packaging shall conform to UFC 6000, National Motor Freight Classification or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), the graphite shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each billet shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

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c. Material trade name and identification.

d. Lot number and date of quality conformance inspection.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The graphite specified herein is intended for use as a component in the Standard Missile Mk 104 Dual Thrust Rocket Motor nozzles.

6.2. **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

c. Whether first article is required (see 3.1 and 6.3).

d. DOD FAR 52.246-15 (see 4.5.1).

6.3. **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quantity of the graphite sufficient to perform all of the inspections specified in 4.3. The first article sample should be packaged and marked as specified in section 5. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 **Subject term (key word) listing.**

Nozzles
Rocket Motor, Dual Thrust, Mk 104
Standard Missile

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6.5 Supersedure information. MIL-G-82841 incorporates the following engineering change proposal (ECP) and specification change notice (SCN):

ECP	SCN
SSS1321 12/9/86	SCN 1 12/9/86

Preparing activity:
NAVY-OS
(Project 9620-N016)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-G-82841	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
GRAPHITE, ATJS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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